RIPHAH INTERNATIONAL UNIVERSITY

DSA LAB

LAB TASK 10

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Subject: Data Structure

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Task 2: Write conditions for:

1. Queue Full condition 2. Queue Empty condition. Explain both in your own words.

1. Queue Full Condition

Explanation:

A queue is full when there is no more space left to insert new elements. In a circular queue, this happens when the rear pointer reaches just before the front pointer because of circular movement. At this point, adding another element would overwrite existing data, so insertion is not allowed. In other words Queue full means no empty space left to store new data.

Condition (Circular Queue):

It has 2 conditions. We can use any:

- Condition 1: (rear == SIZE-1 && front == 0)
- Condition 2: (rear == front - 1)

Code Example (C++)

<u>Condition 1:</u>

```
bool isFull() {
   if (rear == SIZE - 1 && front == 0) {
      return true;
   }
   return false;
}
```

<u>Condition 2:</u>

```
bool isFull() {
   if (rear == front - 1) {
      return true;
   }
   return false;
}
```

Explanation:

1. **rear == SIZE - 1 && front == 0**

   It means rear is at the last index of array and front is at the first index (0)**.**

So the queue looks like:

```
[ X X X]
  ↑     ↑
front   rear
```

No space left so Queue is FULL

2. rear == front - 1

This is the circular condition and it means rear is just behind front

Example:

```
Index:  0   1   2   3   4
       [X] [X] [ ] [X] [X]
              ↑   ↑
            rear front
```

- Here rear = 1 and front = 2
- So: rear == front - 1

Again no space left so Queue is FULL

Return values:

- true means Queue is full
- false means Space is available

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2. Queue Empty Condition

Explanation:

A queue is empty when there are no elements present in it. This means no data is available to remove (dequeue). In a circular queue, this condition occurs when the front pointer does not point to any valid element.

At this point, deletion is not allowed because there is nothing to remove. In other words Queue empty means no data is available in the queue.

Condition (Circular Queue):

It has 1 main condition:

- Condition: (front == -1)

Code Example (C++)

```
bool isEmpty() {
    if (front == -1) {
        return true;
    }
    return false;
}
```

Explanation:

1. front == -1

It means the queue has **no elements** and it is the initial state of the queue. It also happens after deleting all elements

So the queue looks like:

```
[   empty   ]

front = -1
rear  = -1
```

No data present so Queue is EMPTY

After Deletion:

When the last element is removed:

```
front = -1;
rear = -1;
```

So Queue becomes empty again

Return values:

- True means Queue is empty
- False means Queue has elements